

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GPH 222 — Past Questions & Materials
by Renance

Course: GPH 222
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200, 300

This study pack was compiled and restructured by Renance for GPH 222 students at Federal University of Petroleum Resources, Effurun. It brings together 4 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 4 — GPH 222 PEE 324 INTRO TO EXPLORATION GEOPHYSICS

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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DEPARTMENT OF EARTH SCIENCES

SECOND SEMESTER: B.SC. (GEOPHYSICS) 2023/2024 SESSION EXAMINATION

COURSE: GPH 222/PEE 324 (2 Units)- INTRO. TO EXPLORATION GEOPHYSICS

TIME ALLOWED: 2Hrs.

INSTRUCTION: ANSWER ALL QUESTIONS IN SECTION-A (COMPULSORY) AND ANY TWO (2) OTHER QUESTIONS FROM SECTIONS-B.

SECTION A

Q1 (30marks)

- (a) (i) The unit of resistivity is.....and the reciprocal of resistivity IS.....-.....:sssees ree
- (ii) The application of geophysics techniques to geophysical problems are employed extensively
- (iii) Exploration for metallic ore minerals is mainly carried out using what geophysical methods?
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- (iv) The instrument used for measuring local variations In magnetic field strength is called
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can be ee ee ae 1 10 eee Te RT Te Se One Emre,
- (vi) The physical croperty measured in magnetic exploration method iS 0.2... cee eee eer eet
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- (vii) The maanetization that remains on a ferromagnetic rock after the external magnetic fie'd is
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- (viii) Th- e'sctrode cc .figuraton'array that hes ihe meximum cepth o invesngauor is
- (ix) _js also known as in incomoressibility in elastic constant definition'
- (x) vo eeee -serereese sin iS based on known haif-life span of Radioactive elenenrs
- (xi) The veiccity at which a= seismic wave travels through ae material depends
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- (.i}The sum energy of the reflected and transmitted waves equals the.....065. ie,
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- (xiv) The oreportion of energy reflected anc or transmitted are determined/depenas <r
the..... oss sss, . cross the interfece/boundary.
- (xv) Ai critical angie, the refracted wave will travel... A
Jelocity bit s.pak SARA
- (xvi). cieomny sienceat is the distance at which the travel time of reflected wave equals the

travel time of refracted wave. (10 marks)

(1D) (i) State the Zoeppritz equations in terms of acoustic impedance (2 marks)

(ii) State any one (1) application each of seismic reflection and refraction methods (2 marks)

(iii) Define the terms Crossover distance and Critical distance. (2 marks)

(iv) Using diagram only depicts the difference between Acrit and Across (2 marks)

(v) A seismic refraction survey gave the following data for the first arrival times at various distances

from the shot-point to determine the depth to bedrock below

“Offset X(m) | 15 | 30 | 45 | 60 | 75 | 90 | 105 | 120 | 135 | 150 | 165 | 180 |

Time t(ms) | 19 | 29 | 39 | 50 | 59 | 62 | 65 | 68 | 72 | 76 | 78 | 83

Plot t vs X

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(2 marks)

(a) Plot the refraction profile for the first break arrivals

(2 marks)

(b) Calculate the seismic velocities of the respective layers

(c) Calculate the minimum depth to the bedrock/retractor

(iv) Calculate the critical angle of incidence and the crossover distance (using equation) (4 marks)

1

SECTION B

Q3 (20 marks)

(a) State the laws of reflection (2 marks)

(b) Derive the time-distance equation of a P-wave from a source, S, striking a reflector at R and recorded at the detector, D after a reflection from point R with the aid of relevant diagram(s) (6 marks)

(c) The following results were obtained by a FUPRE Student from a seismic reflection data acquired in Lembe Creek in BAYESA State. The vertical two-way travel time, $T_0 = 0.1474$ seconds and the slope of the $t_y - x_g$ Plot, $S = 3.0857 \times 10^{-7} \text{ s/m}^*$. Compute:

(i) The velocity, V to the reflecting interface (2 marks)

(ii) The depth to the reflector Z (2 marks)

(iii) The total travel time, T_x for the wave generated at the Source, S to get to the far end Geophone Receiver, R, 2,500m away from the Source. (2 marks)

(iv) The angle of incidence, θ . (2 marks)

(v) Sketch the time-distance graph obtained from the reflection data (2 marks)

Q4 (20 marks)

(a) What are the measured parameter, calculated parameter and main applications of electrical

resistivity method of geophysical prospecting? (5 marks)

(b) State Ohm's law and show that it can be written in this form $E=pJ$ (6 marks)

(c) Write equations to describe the emissions of α -particle, β -particle and γ -particle by unstable radioactive nucle (3 marks)

(d) Give three (3) properties of radioactive emission and name three (3) instruments used to detect the

emission f6 marks,

_ Q5 (20 marks)

2) (i) Compare and contrast Gravity and Magnetic methods (4 Marks)

(ii) Briefly discuss the Accuracy and Precision of geophysical instruments (4 Marks)

(b) Briefly explain the following:

(i) Coulomb's Law with an example

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(ii) Magnetic Susceptibility a ied

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| Q6 (20 marks)

(a) Write short notes on the following:

(i) Flux - gate Magnetometer

(ii) Geiod (4 Marks)

(4 Marks)

(iii) Calibration of Gravimeter

i (4 Marks)

(b) With the aid of a diagram, show the Schematic section of | end

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NOTE: GRAPH SHEETS TO BE PROVIDED IN THE EXAMINATION HALL.

Paper 2 of 4 — GPH 222 PEE 324 INTRO TO EXPLORATION GEOPHYSICS

Level 200 • Second Semester • PastQuestion

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DEPARTMENT OF EARTH SCIENCES

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Paper 3 of 4 — GPH 222 past question

Level 200 • Second Semester • PastQuestion

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DEPARTMENT OF EARTH SCIENCES

SECOND SEMESTER: 6.SC. (GEOPHYSICS) 2023/2024 SESSION EXAMINATION

COURSE: GPH 222/PEE 324 (2 Units)- INTRO. TO EXPLORATION GEOPHYSICS

: 2Hrs.

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Paper 4 of 4 — GPH 222&PEE 324

Level 300 • Second Semester • PastQuestion

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DEPARTMENT OF EARTH SCIENCES

SECOND SEMESTER: 6.SC. (GEOPHYSICS) 2023/2024 SESSION EXAMINATION

COURSE: GPH 222/PEE 324 (2 Units)- INTRO. TO EXPLORATION GEOPHYSICS

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noe betwee Xerit and Xcross (2 marks)

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